Extremes for metocean: Fundamentals

Philip Jonathan

Lancaster University

IOGP, London, September 2026
(slides at *www.lancs.ac.uk/∼jonathan)*



Motivation for masterclass

Problem set-up

- We think of the ocean environment as a "data generating process"
- Sometimes that process generates extreme values of particular concern
- The only way we can understand the extremes is by direct observation
- We fit an empirical tail model to data we collect
- We use the model to predict how often and how big the extremes will be

What are the issues?

- Which tail model should we fit? And why?
- How do we fit the tail model?
- How do we know we've achieved a good fit? How useful is the model?
- What if the process is non-stationary (ie there are covariates at play)?
- What if we want to fit a multivariate tail model?
- How do we tie this all together for engineering design?

In other words ...

In this workshop, I'm trying to summarising the things I wish I'd known when I started working on extreme value analysis for metocean

What does theory say we should do? How do we check we've done something sensible? How do we handle the more complicated applications? How do I know if the "way I usually do it" is good enough?

It's impossible to discuss everything in an afternoon, but I hope to "sow a few seeds", which will motivate us to do some more reading and thinking, and playing with our favourite Foundation / GenAI / programming tools ☺

Overview of masterclass

- A: Fundamentals
- B: Covariates
- C: Multivariate
- D: covXtreme
- E: Recent developments & take-aways

Overview of A: Fundamentals

- Motivating example
- Tail models commonplace in engineering
- Statistically-motivated tail models
- Fitting a tail model
- Key issues with GPD fitting of peaks-over-threshold
- Uncertainty quantification
- Basic statistical model fitting tools

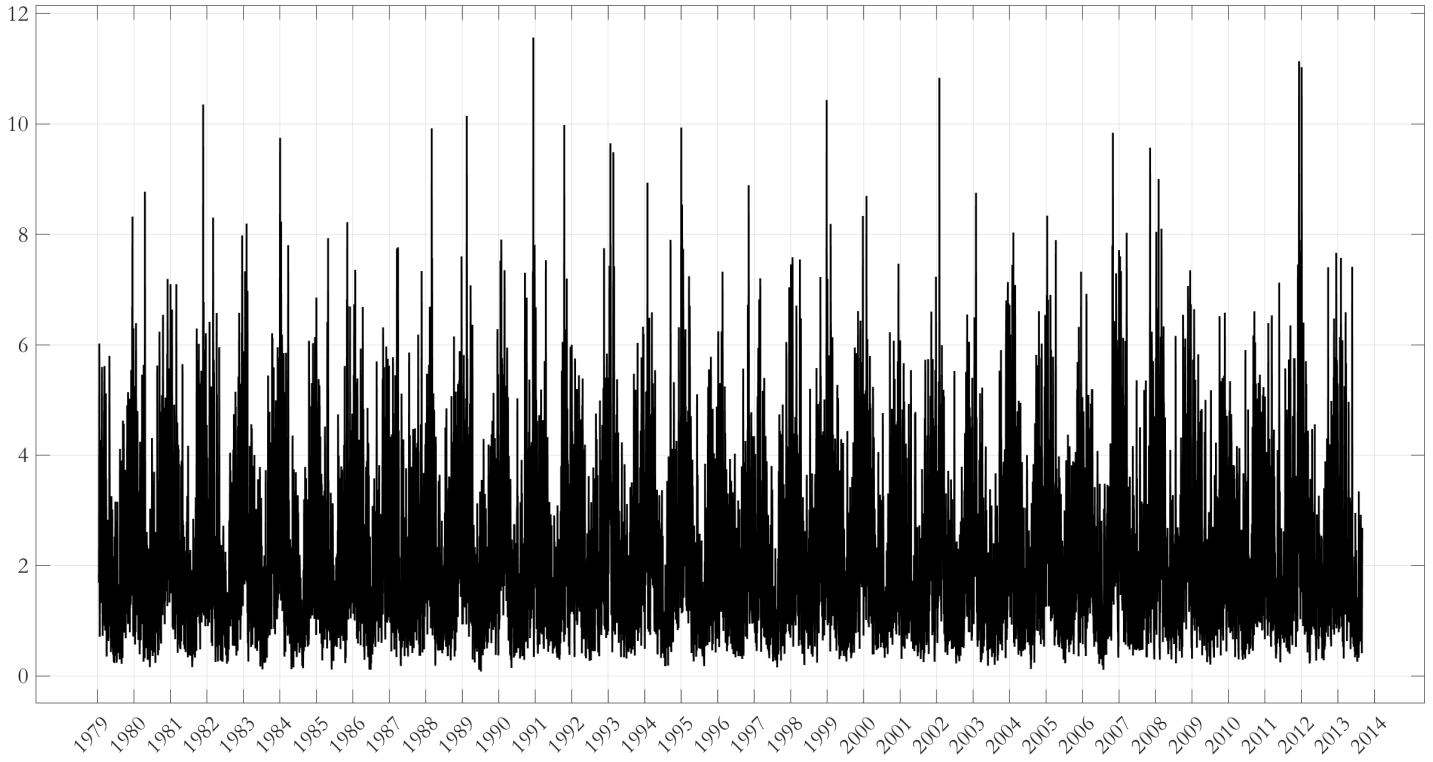
Sanity check

- All models are wrong, some models are useful
- George Box, `http://en.wikipedia.org/wiki/GeorgeE.P.Box`

- Consistent balance of engineering and statistical insights

Motivating application

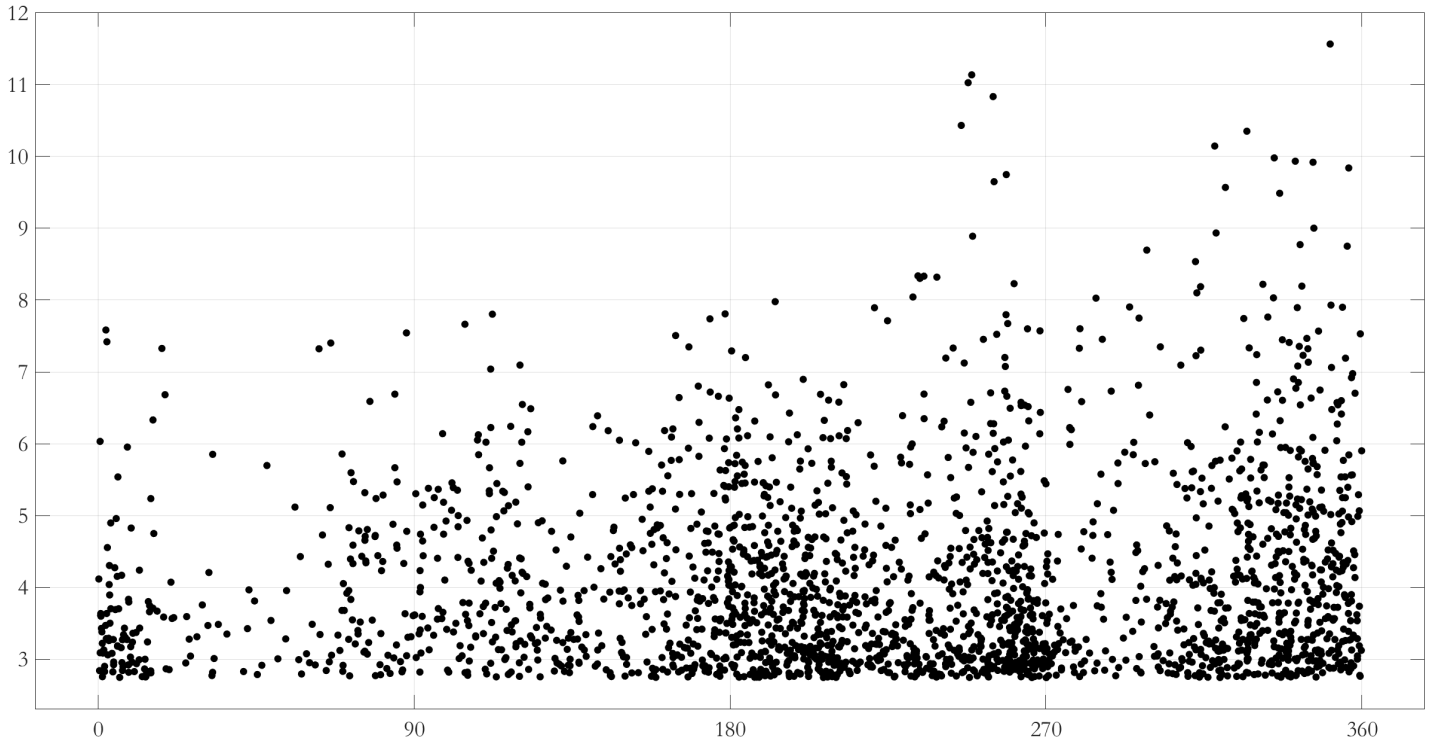
- 3-hour sea state H_S from a central North Sea location
- Data available from covXtreme repository



Time-series of H_S

Motivating application

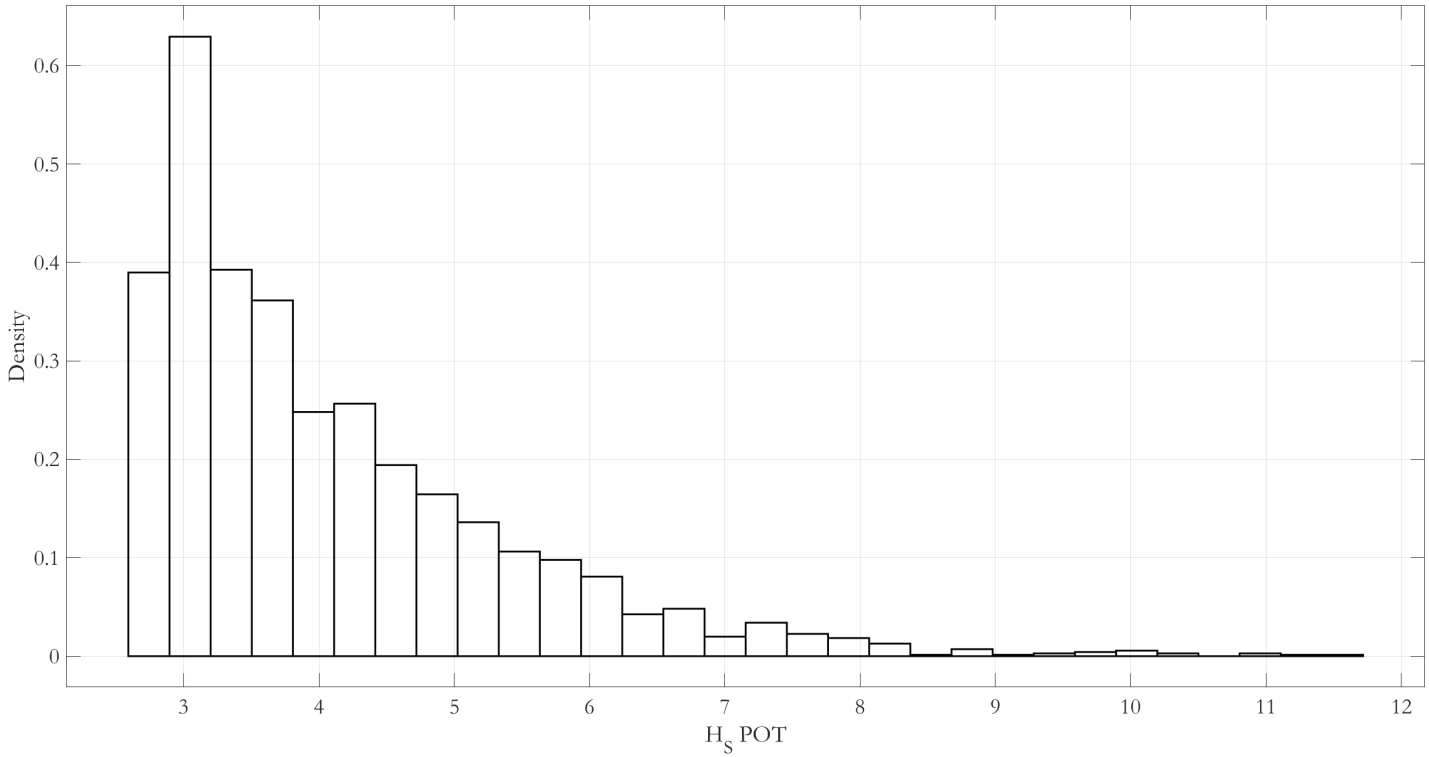
- Peak-picking algorithm used to isolate storm peaks H_S^{sp} over threshold



H_S POT on direction

Motivating application

- Density of H_S^{sp}



Density of H_S POT

Typical engineering approach

- Some variant of "Weibull" distribution
- For random variable X, e.g. H_S^{sp}

$$\mathbb{P}(X \leq x; \lambda, k) = F(x; \lambda, k) = \begin{cases} 1 - e^{-(x/\lambda)^k} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- Shape $k > 0$, scale $\lambda > 0$
- There is no limit to how big X can get

- Widely used in science and engineering
  - Changing k (and λ) gives flexibility
  - Tail of density decreases for large x
- But there is no "fundamental" reason for using this distribution

Understanding the statistical approach

Degenerate distribution of block maxima

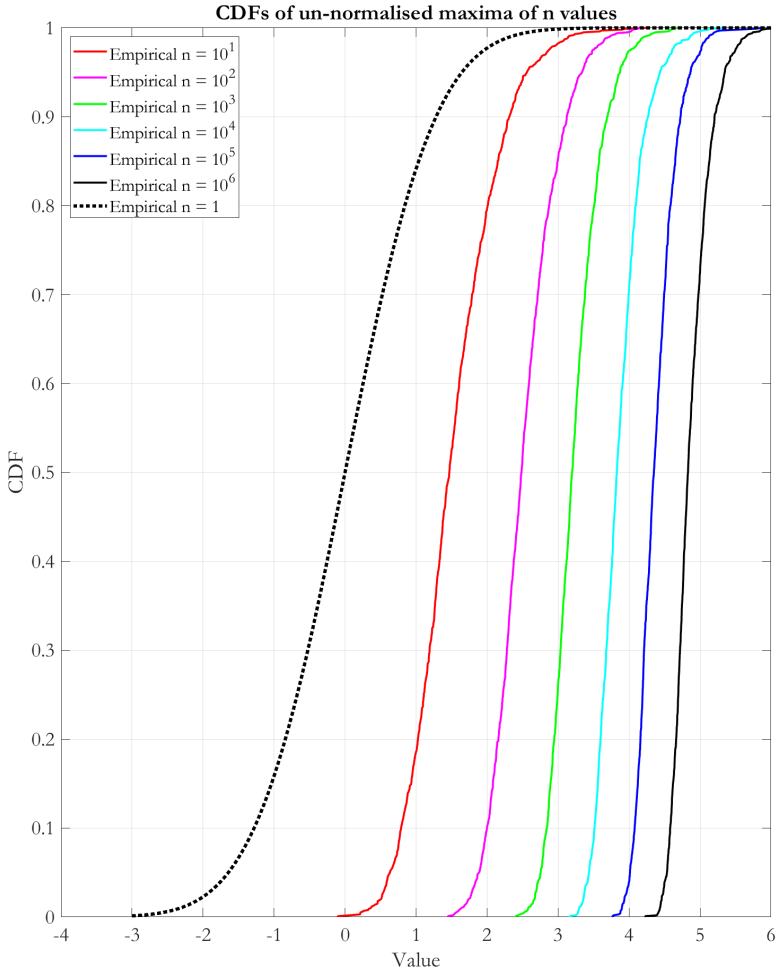
- $F(X) = \mathbb{P}(X \leq x)$, cumulative distribution function (cdf)
- $M_n = \max_i\{X_i\}$, block maximum over the n values X_i, $i = 1, 2, ..., n$
- $\mathbb{P}(M_n \leq x) = [\mathbb{P}(X \leq x)]^n$, cdf of maximum
- As $n \uparrow \infty$, $\mathbb{P}(M_n \leq x)$ becomes degenerate ($= 0$ everywhere except at the maximum value of X, x^*)
- What do we do to make $\mathbb{P}(M_n \leq x)$ useful?

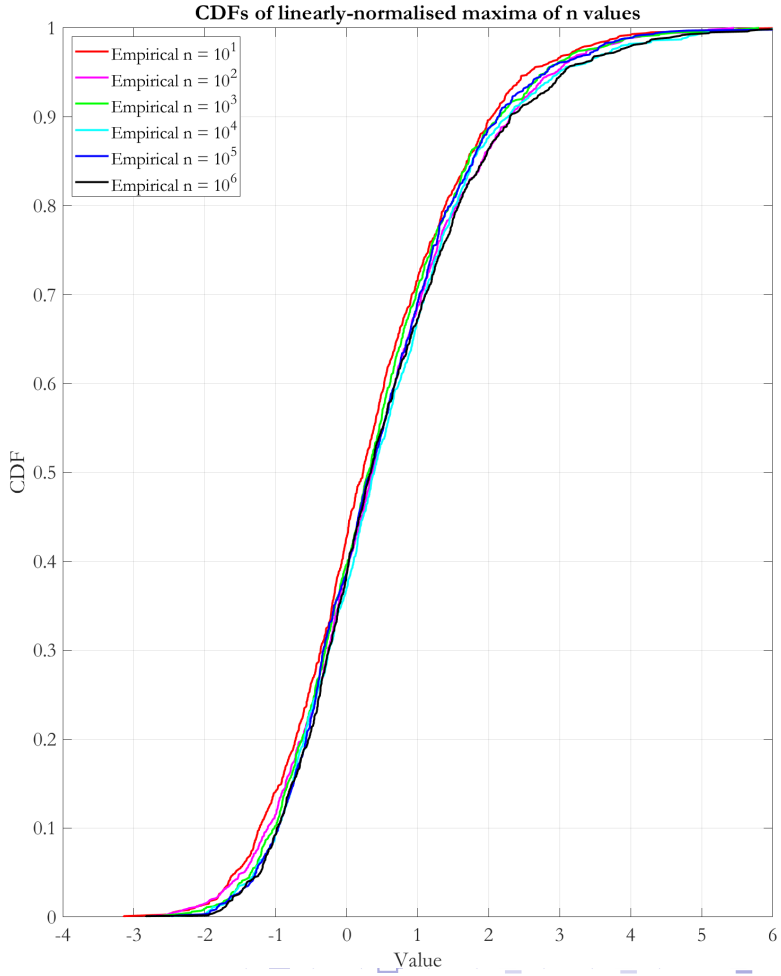
Non-degenerate distribution of shifted and scaled block maxima

- Try shifting and scaling the random variable to make its tail more stable
- $Y_n = a_n^{-1}(max_i\{X_i\} - b_n)$
- $\mathbb{P}(Y_n \leq y) = [\mathbb{P}(X \leq b_n + a_n y)]^n$
- As $n \uparrow \infty$, $\mathbb{P}(Y_n \leq y)$ is well behaved once we find good a_n and b_n
- The tail of the distribution of Y_n is max-stable!
- a_n and b_n are functions of block size n (examples coming!)

Understanding the statistical approach

- Left: distribution $[\mathbb{P}(X \leq x)]^n$ of un-normalised Gaussian maxima becomes degenerate
- Right: distribution $[\mathbb{P}(X \leq a_n y + b_n)]^n$ of linearly-normalised maxima converge to standard Gumbel





In other words ...

If we look at the "raw" distribution of the "block maximum", we get something silly as block size gets big: the distribution becomes a step function. This is USELESS for predictions of what might happen with even bigger blocks!

But if we simply "shift and scale" or "normalise" the block maximum, we get something REMARKABLE happening: the distribution of the block maximum converges to a COMMON UNIQUE form, characterised by just ONE PARAMETER ξ. The limit distribution is the GEV

This might feel weird, but in reality it's a lot like the central limit theorem, which says that the BLOCK MEAN, when appropriately shifted and scaled, converges to the standard Gaussian distribution

Understanding the statistical approach

The non-degenerate limiting distribution is unique

- $Y_n = a_n^{-1}(max_i\{X_i\} - b_n)$
- $\mathbb{P}(Y_n \leq y) = [\mathbb{P}(X \leq b_n + a_n y)]^n$

Generalised extreme value distribution (GEV)

$$Pr(Y_n \leq y) \quad \rightarrow \quad \exp\{-(1+\xi\frac{y-\mu}{\sigma})^{-1/\xi}\} \text{ as } n \rightarrow \infty \text{ for } \xi \neq 0, y > \mu$$

$$(\quad \rightarrow \quad \exp\{-\exp(-\frac{y-\mu}{\sigma})\} \text{ when } \xi = 0)$$

- Location $\mu \in \mathbb{R}$, scale $\sigma > 0$, shape $\xi \in \mathbb{R}$

Maximum domain of attraction (MDA)

- A distribution whose (linearly normalised) maxima converges, belongs to the MDA of a max-stable distribution (for some ξ)
- We call these distributions "nice"; most continuous distributions are nice
- Once we find ξ for one choice of a_n and b_n, it must be the same for all choices of a_n and b_n

- The Weibull distribution converges to GEV with:
  - $\xi = 0$
  - $a_n = \frac{\lambda}{k}(\log n)^{1/k} - 1$
  - $b_n = \lambda(\log n)^{1/k}$ ($= F^{-1}(1 - 1/n)$ in general)

- Note: this theory is analogous to central limit theorem. There is nothing mysterious here!

Unified von Mises condition

- For any nice distribution, the ultimate value of GEV ξ is given by

$$\lim_{x \to x^*} \frac{d}{dx}\left(\frac{1}{g(x)}\right) = \xi, \quad g(x) = \frac{f(x)}{1 - F(x)} = \text{hazard}$$

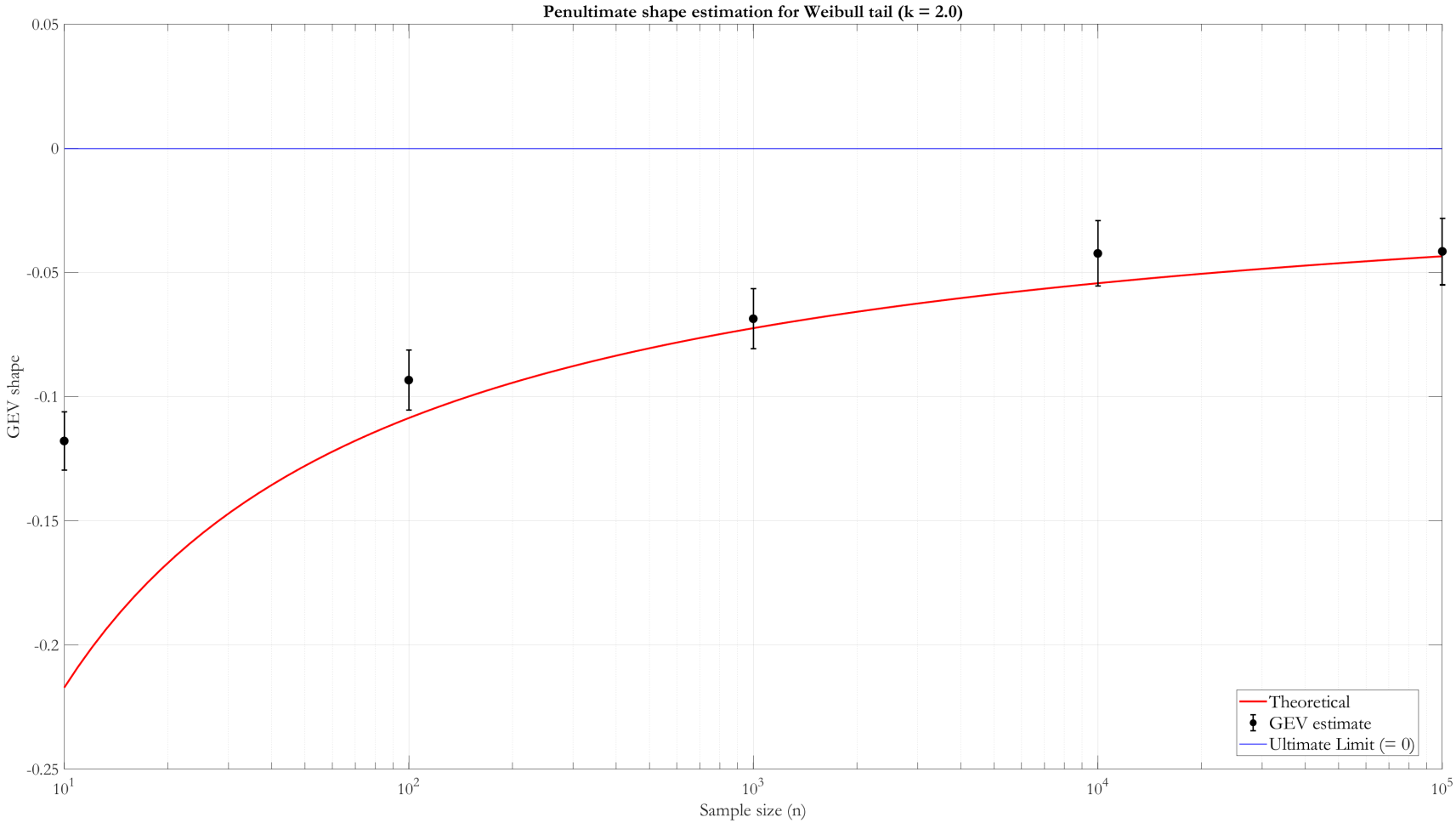
- The "penultimate approximation" ξ_n of ξ, for sample size n can be estimated using

$$\xi_n = \frac{d}{dx}\left(\frac{1}{g(x)}\right)_{x=b_n} \approx \frac{1-k}{k \ln n} \text{ for Weibull}$$

- Penultimate approximate gives better fit for samples from distributions which converge slowly to ultimate tail
- For finite Weibull sample ($n < \infty$) with $k > 1$ (e.g. $k \approx 2$, Rayleigh), we have $\xi_n < 0$, and we expect to fit a shorter tail with $x^* < \infty$
  - We should fit a GEV, but expecting $\hat{\xi} < 0$
  - We need to adjust our estimate of ξ_n for larger return value calculations!

Penultimate approximation for Weibull tail

- Weibull sample size n for $k = 2$ (Rayleigh)
  - Penultimate $\xi_n < 0$: we expect to fit a shorter tail expecting $\hat{\xi} < 0$
  - Hence expect finite upper end point $x^* < \infty$
  - Adjust estimate $\xi_{n'}$ to use in return value calculations when $n' > n$



In other words ...

The "MDA stuff" is just the formal way of saying a distribution is "nice". Distributions we meet in environmental and engineering applications are typically "nice". So we can expect that the distribution of the block maximum (for big n) looks like a GEV

The distribution of the block maximum from the "engineering Weibull" actually converges to a Gumbel distribution (i.e. a GEV with $\xi = 0$). So there is no upper limit to how large the block maximum can get. This doesn't make physical sense

However, convergence for the "engineering Weibull" block maximum is really slow. So if we simulate a sample of Weibull data, then fit a GEV, we will get a NEGATIVE shape parameter estimate. We can actually work out from the maths how this estimate will vary as a function of sample size n. As n gets really big, the shape parameter estimate converges to zero

So if we knew we had Weibull data, we could adjust the negative shape parameter to account for return period. But of course in practice, we don't KNOW we have Weibull data

What about POT?

- X has distribution function F in MDA of GEV(ξ)
- If n is large enough, $F^n(a_n x + b_n) \approx \exp(-\left(1 + \xi \frac{x-\mu'}{\sigma'}\right)^{-\frac{1}{\xi}})$ (GEV tail)
- So $F^n(x) \approx \exp(-\left(1 + \xi \frac{x-(a_n\mu'+b_n)}{\sigma' a_n}\right)^{-\frac{1}{\xi}}) = \exp(-\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-\frac{1}{\xi}})$ say
- Then $n \log F(x) \approx -\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-\frac{1}{\xi}}$ (log both sides)
- $\mathbb{P}(X > x) = 1 - F(x) \approx \frac{1}{n}\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-\frac{1}{\xi}}$ ($\log y \approx -(1-y)$) for $y \approx 0$)
- $\mathbb{P}(X > x | X > u) = \frac{1-F(x)}{1-F(u)} \approx \left(1 + \xi \frac{x-u}{\tilde{\sigma}}\right)^{-\frac{1}{\xi}}$ (re-arrangement)
- where $\tilde{\sigma} = \sigma + \xi(u - \mu)$ is the adjusted scale; common ξ for GEV and GP!

Generalised Pareto distribution (GP or GPD)

$$\mathbb{P}(X > x | X > u) = \left(1 + \xi \frac{x-u}{\sigma}\right)^{-\frac{1}{\xi}} \qquad \xi \in \mathbb{R}, \sigma > 0, x > u, x \leq x^*$$

- Block maxima from nice distributions $\to$ GEV as $n \to \infty$
- Threshold exceedences from nice distributions $\to$ GP as $u \to x^*$

Recovering GEV from GP

- X has a nice distribution function F, conditional tail $F_u(x) = \frac{F(x)}{1-F(u)}$
- Assume occurrence rate of exceedences over threshold u is Poisson-distributed

$$\begin{aligned}
\mathbb{P}(\text{maximum in period } \leq x | x > u) &= \sum_{k=0}^{\infty} \mathbb{P}(k \text{ storms in period}) F_u^k(x) \\
&= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \exp(-\lambda) F_u^k(x) \\
&= \exp\{-\lambda(1 - F_u(x))\}
\end{aligned}$$

- But threshold exceedences are GP, so $1 - F_u(x) = \left(1 + \xi \frac{x-u}{\tilde{\sigma}}\right)^{-\frac{1}{\xi}}$, so

$$\begin{aligned}
\mathbb{P}(\text{maximum in period } \leq x | x > u) &= \exp\{-\lambda \left(1 + \xi \frac{x-u}{\tilde{\sigma}}\right)^{-\frac{1}{\xi}}\} \\
&= \exp\{-(1 + \xi \frac{x-\mu}{\sigma})_+^{-1/\xi}\} \quad \text{(GEV)}
\end{aligned}$$

- when λ set to $\left(1 + \xi \frac{u-\mu}{\sigma}\right)^{-\frac{1}{\xi}}$ for $\tilde{\sigma} = \sigma + \xi(u - \mu)$ wlog; ξ unchanged

Take-aways

- Statistics suggests we should fit tails of distributions with GEV and GP
  - Motivation for GEV and GP is asymptotic theory
  - Can only justify fitting GEV and GP when we are in the tail
  - Threshold exceedences are GP if u large enough
  - Block maxima are GEV if n large enough

- "(Engineering) Weibull" is a restricted choice of distribution for modelling corresponding to $\xi = 0$ (and slow convergence)
  - Physics (e.g. constraints like Miche for $H_{\max}$ and water depth for H_S) tells us that Weibull cannot always be correct
  - Weibull might be easier to fit to data (since it is more restricted), but this doesn't necessarily make it better

- So how do we proceed?
  - Simulation studies for known data generating models
  - Understand characteristics of models under misspecification
  - Consistent balance of engineering and statistical insights

In other words ...

Provided the underlying distribution is "nice", the distribution of the block maximum for large n looks like a GEV, and the distribution of a threshold exceedance for large threshold u looks like a GP. You can use one result to prove the other

Weibull (and Gaussian) distributions are complicated, because the distribution of the extremes they produce converges very slowly to the limit

Fitting a Weibull distribution isn't "particularly wrong", but it will likely lead to bias in return values; we would effectively assume that the tail of the distribution is the same everywhere. Theory suggests we should fit GEV or GP; but these may also lead to bias (e.g.) for small sample sizes, or if we ignore covariates, or we set n or u too small, or ...

The best solution is to use the methodology which minimises our losses / maximises utility over time ... all models are wrong, some models are useful

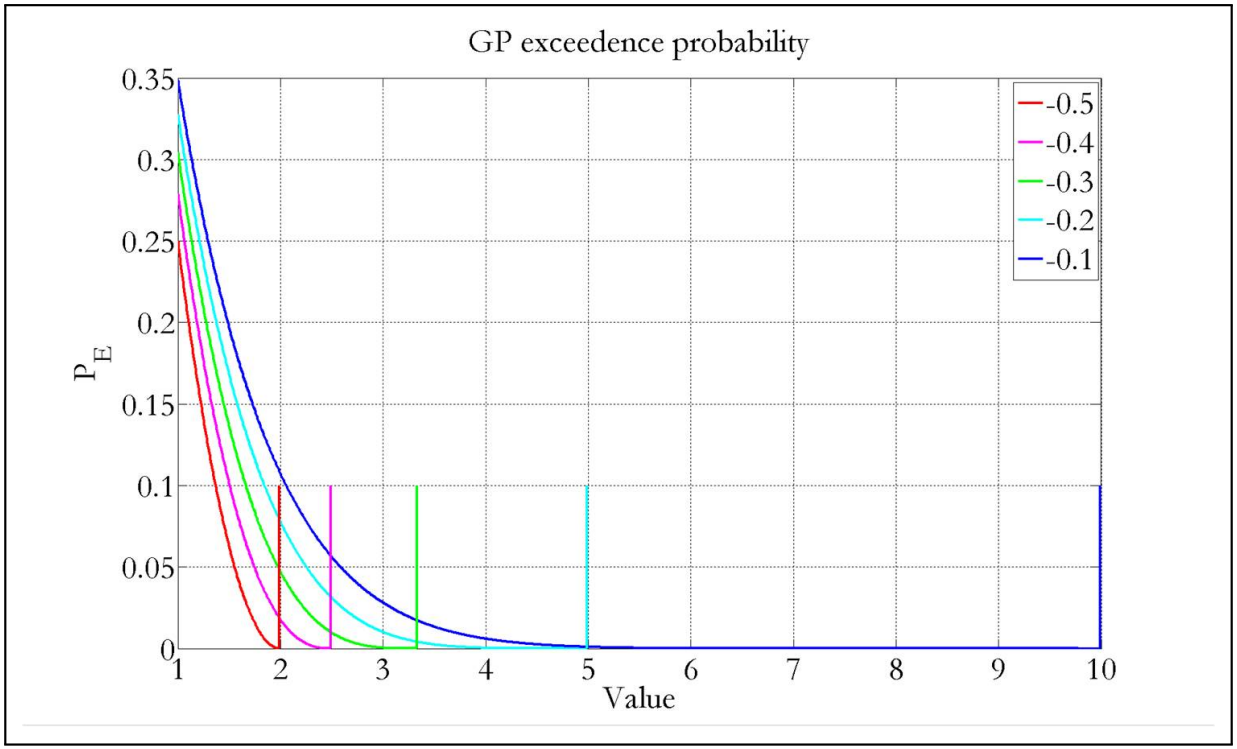
GP: effect of ξ

- $\mathbb{P}(X > x | X > u) = (1 + \xi \frac{x-u}{\sigma})^{-\frac{1}{\xi}}$
- If $\xi < 0$, there is a finite upper end-point x^* which cannot be exceeded
- If $\xi \geq 0$, the upper end-point $x^* = \infty$

- For ocean waves, observation and physics suggests that x^* is finite. e.g Miche:
  - $\frac{1}{2} k_L H_{MAX} = 0.14\pi \tanh(k_L d)$
  - In deep water, Taylor expansion yields $k_L H_{MAX} \approx 0.88$ limit
  - In shallow water, Taylor expansion yields $\frac{H_{MAX}}{d} \approx 0.88$ limit

- Weibull distribution has the upper end-point $x^* = \infty$

GP: effect of $\xi < 0$



- $x^* = u - \sigma/\xi$ for $\xi < 0$
- Assuming $u = 0, \sigma = 1$ in the figure, and $\mathbb{P}_E = 1 - F_u$
- As $\xi \uparrow 0$, then $x^* \uparrow \infty$

GP: threshold stability of ξ

- Consider changing threshold from u to v, $v > u$
- Then $\mathbb{P}(X > x | X > u) = (1 + \xi \frac{x-u}{\sigma})^{-\frac{1}{\xi}}$

$$\begin{aligned}
\mathbb{P}(X > x | X > v) &= \frac{\mathbb{P}(X > x)}{\mathbb{P}(X > v)} \\
&= \frac{\mathbb{P}(X > x | X > u)}{\mathbb{P}(X > v | X > u)} \\
&= \frac{(1 + \xi \frac{x-u}{\sigma})^{-\frac{1}{\xi}}}{(1 + \xi \frac{v-u}{\sigma})^{-\frac{1}{\xi}}} \\
&= (1 + \xi \frac{x - v}{\sigma + \xi(v - u)})^{-\frac{1}{\xi}}
\end{aligned}$$

- Conditional tail is still GP
- ξ is unchanged, σ varies linearly with gradient ξ aafo threshold
- This is a useful fit diagnostic!

GP: return values

Distribution of annual maximum, A

- $X|(X > u) \sim F_u(x) = \text{GP}(\xi, \sigma, u) = 1 - (1 + \xi \frac{x-u}{\sigma})^{-\frac{1}{\xi}}$
- $F_{A;u}$ is the (conditional) distribution of the annual maximum event A

$$F_{A;u}(x) = \exp\{-\lambda(1 - F_u(x)\}$$

P-year return value, x_P

$$1 - \frac{1}{P} = \exp\{-\lambda(1 - F_u(x_P))\}$$

- λ is the expected number of exceedences $X > u$ per annum.

Distribution of P-year maximum, A_P

$$F_{A_P;u}(x) = \exp\{-P\lambda(1 - F_u(x)\}$$

Equivalent expressions for GEV

In other words ...

ξ tells us all the important stuff about the limit distribution. $\xi < 0$ means there is a "finite upper end point" which cannot be exceeded. $\xi > 0$ means "heavy tails" which are unlikely in practice for typical metocean applications (just from energy considerations)

If a GP seems to fit well at one threshold u, it should also work well WITH THE SAME VALUE OF ξ at higher thresholds. So plotting estimated ξ against u and looking for stability is a great diagnostic! You can also check for stability of return value x_p against u

If u is too small, there's bias because we are not "asymptotic enough", and the GP doesn't describe the data well. If u is too high, the estimate of ξ will be very noisy because we have just a small sample to estimate it! The "Goldilocks" region is in the middle

In my experience, for storm peak H_S, we have $\xi \in [-0.4, 0)$, nearer zero when e.g. directions correspond to long fetches for extratropical storms, or peak hurricane directions

In practice: threshold choice

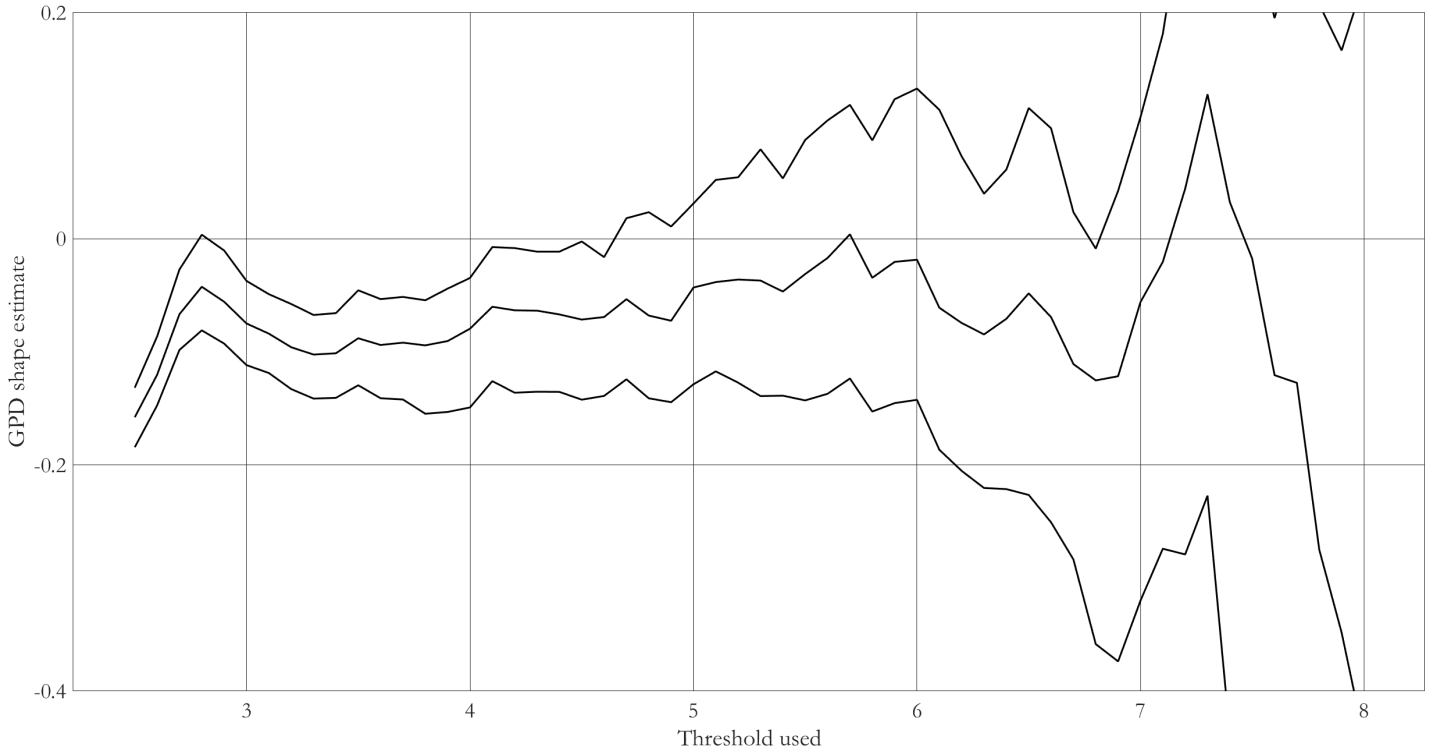
- We're fitting threshold exceedences using GP tail. What do we need to consider?
- Choice of threshold, u
  - If GP fits well, estimated ξ is constant as we vary u
  - Also check if estimated return value x_p is constant with u
  - Do we need to choose just one threshold? What about model averaging?
- PP and QQ plots (more later)
- Uncertainty in estimates
  - As sample size $\downarrow$, uncertainty in estimated parameters and $x_p \uparrow$
  - Often good to look at uncertainty of estimated x_p
  - Use bootstrap resampling

In practice: bootstrapping

- A data-driven way to estimate uncertainty, making minimal assumptions
- The procedure is:
  - Estimate the quantity you're interested in, using the original sample; let's say it's a return value x_P
  - Draw a new sample of the same size as the original by resampling from the original with replacement
  - Re-do the analysis with the new sample, and obtain a bootstrap estimate $x_P^{(1)}$ for the return value
  - Do the same repeatedly, for a large number (≈ 250) of bootstrap resamples, to obtain a set $\mathcal{B}$ of return value estimates $\mathcal{B} = \{x_P^{(1)}, x_P^{(2)}, x_P^{(3)}, ..., x_P^{(250)}\}$
  - Use $\mathcal{B}$ to estimate the uncertainty of the original estimate x_P
- Standard practice in the statistics community

In practice: stability of shape estimate, $\hat{\xi}$

- North Sea data
- $\hat{\xi}$ should be stable with u if GP appropriate
- Plot shows $\hat{\xi}$ for actual sample (middle line)
- Also shows 95% bootstrap uncertainty band from resampling
- $u \in [3, 6]$ yields reasonably stable model



Stability of $\hat{\xi}$

In practice: cross-validation

- A data-driven way of assessing predictive performance
- Hence: a way of selecting the optimal value for a hyper-parameter
  - e.g. GP threshold u
- Or: a way of deciding which of competing models is best
  - e.g. GP or Weibull
- The procedure is:
  - Partition the sample into k groups
  - Estimate model using $k - 1$ of the groups, keeping the "withheld" group as a prediction (or "test") set
  - Assess predictive performance on the prediction set
  - Repeat until all k groups have been used exactly once for prediction
  - Estimate overall "k-fold" predictive performance for full sample
  - If necessary, choose the hyper-parameter or model that maximises predictive performance
- Standard practice in machine learning and computational statistics

In practice: model fitting

Maximum likelihood estimation

- $X|X > u \sim F_u(\xi, \sigma) = GP(\xi, \sigma)$, with density $f(x; \xi, \sigma)$
- $f(x; \xi, \sigma) = d(F_u(x))/dx$
- Observation x_i drawn from $GP(\xi, \sigma)$ has "likelihood" $f(x_i; \xi, \sigma)$
- Likelihood of whole sample $(x_1, x_2, x_3, ..., x_n)$ is $\mathcal{L}(\xi, \sigma)$

$$\mathcal{L}(\xi, \sigma) \stackrel{\triangle}{=} \prod_{i=1}^{n} f(x; \xi, \sigma)$$

- Choose estimates of ξ and σ which maximise sample likelihood
- Usually have to estimate ξ and σ numerically

- Great coupled with cross-validation for hyper-parameter optimisation
- Equivalent to least squares for Gaussian samples
- Alternatives: method of moments for small samples

In practice: hyper-parameter and model selection

- We have a fiddle factor θ that we can't estimate easily. Why?
  - θ could be the GP threshold u
  - θ could switch between a GP tail ($\theta = 0$ say) and Weibull ($\theta = 1$ say)
- Use maximum likelihood estimation and cross-validation together

- Partition sample S into k groups $S_1, S_2, ..., S_k$
- Evaluate optimal predictive likelihood for hyper-parameter choice θ

$$\mathcal{L}^{\text{Opt}}_{\text{Pred}}(S;\theta) = \max_{\xi,\sigma} \prod_{j=1}^{k} \mathcal{L}_{\text{Pred}}(S_j;\xi,\sigma,S_{-j},\theta)$$

- $\mathcal{L}_{\text{Pred}}(S_j;\xi,\sigma,S_{-j},\theta)$ is the predictive likelihood of group S_j for a model estimated using S_{-j} (i.e. all data except S_j), parameters ξ and σ, and hyper-parameter θ

- Choose θ maximising $\mathcal{L}^{\text{Opt}}_{\text{Pred}}(S;\theta)$: $\theta^{\text{Opt}} = \arg\max_\theta \mathcal{L}^{\text{Opt}}_{\text{Pred}}(S;\theta)$
- A data-driven way to choose optimal threshold for GP, and even tail model form

In practice: penalised maximum likelihood

- What if parameters ξ and σ are subject to other constraints?
- Suppose we can write the "badness" of constraint violation as $C(\xi, \sigma)$, with $C = 0$ meaning that constraints are not violated
- We can perform maximum likelihood estimation with constraints

$$\begin{aligned} \mathcal{L}^*(\xi, \sigma) &= \mathcal{L}(\xi, \sigma) \exp\{-C(\xi, \sigma)\} \\ \text{or } \log(\mathcal{L}^*(\xi, \sigma)) &= \log(\mathcal{L}(\xi, \sigma)) - C(\xi, \sigma) \end{aligned}$$

- So we just maximise $\mathcal{L}^*$ in place of $\mathcal{L}$ over parameters
- Principled way of imposing constraints on estimates
- An intuitive Bayesian interpretation: "posterior $\propto$ likelihood $\times$ prior"
- Reverts to maximum likelihood estimation if $C(\xi, \sigma) = 0$
- Key to fitting flexible covariate models (more later)

In practice: model averaging

- Can't decide which threshold to use, or whether we use GP or Weibull?
- Construct an ensemble of models, and use them all in decision making!
- Estimate m models $(\mathcal{M}_1, \mathcal{M}_2, ..., \mathcal{M}_m)$
- Then we might estimate m return values $(x_{P;\mathcal{M}_1}, x_{P;\mathcal{M}_2}, ..., x_{P;\mathcal{M}_m})$, and use the full set of return value estimates for decision making, e.g.

$$x_P^* = \frac{1}{m} \sum_{j=1}^{m} x_{P;\mathcal{M}_j}$$

- Or $f^*(z) = \frac{1}{m} \sum_{j=1}^{m} f_{\mathcal{M}_j}(z)$ for "predictive" density of quantity z
- Principled model combination with intuitive Bayesian interpretation
- Weight models according to our beliefs about their relative performance
- Key to incorporating threshold and sampling uncertainty in covariate models (more later)

In practice: Bayesian inference

- Need a systematic way of incorporating prior knowledge?
- Specify a prior density $f(\theta)$ for parameters ($\theta = \xi, \sigma, ...$)
- Combine with sample likelihood $\mathcal{L}(\theta) \triangleq f(\text{Data}|\theta)$
- Estimate posterior density $f(\theta|\text{Data})$ for parameters

$$f(\theta|\text{Data}) = \frac{f(\text{Data}|\theta)f(\theta)}{\int_{\theta'} f(\text{Data}|\theta')f(\theta')d\theta'} \propto f(\text{Data}|\theta)f(\theta) = \text{Likelihood} \times \text{Prior}$$

- Calculate posterior predictive estimates for any quantity of interest
  - Posterior predictive estimate e.g. of return value $q(\theta)$
  - Posterior predictive density e.g. of sea state H_S or individual $H_{\max}$, $f(z|\theta)$

$$\begin{aligned} q_{\text{PP}} &= \int_\theta q(\theta)f(\theta|\text{Data})d\theta \\ f(z|\text{Data}) &= \int_\theta f(z|\theta)f(\theta|\text{Data})d\theta \end{aligned}$$

- Posterior predictive estimates are just "special" model averages
  - Weighted, with weights given by posterior density of parameters

In other words ...

"Maximum likelihood estimation" (MLE) is the standard way of fitting models in statistics; loads of background theory. For small samples, other approaches sometimes have better characteristics

Cross-validation and bootstrapping are simple ideas which re-use the sample of data in clever ways to estimate uncertainties, hyper-parameters and predictive performance. Ideas are easy to communicate to others

Bayesian inference is the best approach, because it incorporates our "prior knowledge" in making estimates, and gives a complete description of uncertainty. It is more fiddly to implement and get right, but standard tools are pretty bullet-proof. Penalised likelihood methods are a "half-way house"

Model averaging is a great way of combining lots of models to make inferences, with a natural Bayesian interpretation. Trivial to implement

Everyone has access to PYTHON. No excuses for not incorporating uncertainty in inferences. Start with maximum likelihood, cross-validation and bootstrapping; then "graduate" to Bayesian methods ☺

Books worth getting

- Coles [2001]: Best readable introduction to extreme value analysis
- Pawitan [2001]: Readable introduction to applied statistical modelling
- Davison [2003]: Best coverage of applied statistical modelling, with extremes examples

References

S. Coles. *An introduction to statistical modelling of extreme values*. Springer, Berlin, 2001.

A. C. Davison. *Statistical models*. Cambridge University Press, Cambridge, UK, 2003.

Y. Pawitan. *In all likelihood: statistical modelling and inference using likelihood*. Clarendon Press, Oxford, 2001.